Executive Summary

Project Title

Predictive Modelling and Risk Scoring for Bank Customer Churn

Overview

Customer churn is one of the major challenges faced by banking institutions, as losing existing customers directly impacts revenue and long-term business growth. This project aims to develop a machine learning-based predictive model that identifies customers who are likely to leave the bank. By predicting customer churn in advance, banks can implement targeted retention strategies and improve customer satisfaction.

Objectives

- Predict customer churn using machine learning techniques.
- Analyse customer behaviour through Exploratory Data Analysis (EDA).
- Compare multiple machine learning algorithms.
- Identify the key factors influencing customer churn.
- Support data-driven customer retention strategies through risk scoring.

Dataset

The project utilizes the **European Bank Customer Churn Dataset**, consisting of **10,000 customer records** and **14 attributes**. The dataset includes demographic, financial, and behavioural information such as Credit Score, Geography, Gender, Age, Balance, Number of Products, Active Membership, Estimated Salary, and the target variable **Exited**, which indicates whether a customer has churned.

Methodology

The project followed a structured machine learning workflow:

- Data Collection
- Data Preprocessing
- Exploratory Data Analysis (EDA)
- Feature Engineering
- Model Development
- Model Evaluation

Three machine learning algorithms were implemented:

- Logistic Regression
- Decision Tree
- Random Forest

Results

The performance of the developed models was evaluated using classification accuracy.

Model	Accuracy
Logistic Regression	81.10%
Decision Tree	77.45%
Random Forest	**86.60%**

Among the evaluated models, **Random Forest** achieved the highest accuracy of **86.6%** and was selected as the final predictive model. Feature importance analysis indicated that **Age**, **Estimated Salary**, **Credit Score**, **Balance**, and **Number of Products** were the most influential variables affecting customer churn.

Business Impact

The proposed predictive model enables banks to identify customers at high risk of churn before they discontinue their services. This supports proactive retention strategies such as personalized offers, improved customer engagement, and targeted marketing campaigns. The project demonstrates how machine learning can enhance business decision-making and improve customer retention in the banking sector.

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- Jupyter Notebook
- Streamlit

Conclusion

This project successfully demonstrates the application of predictive analytics and machine learning in customer churn prediction. The Random Forest model provided the best predictive performance, making it suitable for customer risk scoring and retention planning. The findings highlight the value of data-driven decision-making in the banking industry and provide a practical framework for reducing customer churn through predictive modeling.